

AI-Powered Customer Support & Service Automation

Project Team

Name	Role	Contribution
Ali Hussain Shaik	Team Member	Developed custom objects, validation rules, record-triggered flows, Prompt Builder configuration, documentation, and testing.
Manohar Renangi	Team Member	Developed data model, custom fields, Lightning App configuration, object relationships, and testing.
Rangineni Vamsi	Team Member	Configured flows, queues, reports, dashboards, and assisted with system testing.

College Name: Audisankara College Of Engineering and Technology

Project Description

The **AI-Powered Customer Support & Service Automation** system is a Salesforce-based application developed to streamline and automate customer support operations for an international electronics and gadgets company. The application provides a centralized platform to manage customers, merchandise, orders, knowledge articles, and customer support cases while improving operational efficiency through automation and artificial intelligence.

The system enables support teams to efficiently create, monitor, and resolve customer issues related to product defects, warranty claims, order tracking, delivery concerns, and general product inquiries. Using Salesforce Flow Builder, routine business processes such as case status updates and priority-based case handling are automated, reducing manual effort and improving response time.

To enhance customer service quality, Salesforce Prompt Builder is integrated to generate AI-powered summaries for customer support cases. These summaries help support agents quickly understand customer issues and provide consistent, accurate responses. Validation rules ensure data accuracy by enforcing mandatory business requirements before records are saved, while support queues organize customer requests for efficient assignment and processing.

The application also includes a customized Lightning App that provides users with a single workspace to access all project modules, including Customer, Merchandise, Orders, User Case, and Knowledge objects. Reports and dashboards offer valuable insights into customer support performance and business operations.

Overall, the AI-Powered Customer Support & Service Automation system demonstrates how Salesforce Service Cloud, Flow Automation, Prompt Builder, and Agentforce technologies can be combined to deliver an intelligent, scalable, and secure customer support solution that improves productivity, enhances customer satisfaction, and reduces operational complexity.

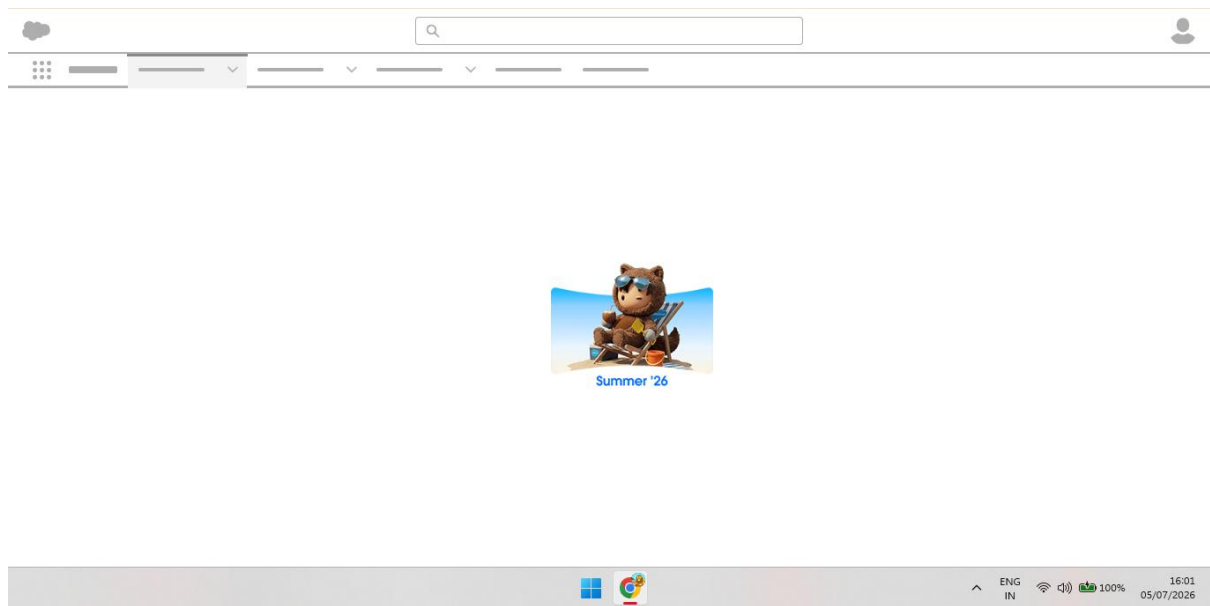
Key Features

- Customer Management using custom Salesforce objects.
 - Merchandise management with warranty and product details.
 - Order management and order tracking.
 - User Case management for customer support requests.
 - AI-generated case summaries using Salesforce Prompt Builder.
 - Record-Triggered Flow automation for case processing.
 - Validation rules to maintain data accuracy and consistency.
 - Support queue management for efficient case assignment.
 - Lightning Application for centralized navigation.
 - Reports and dashboards for monitoring customer support activities.
 - Secure data access through Salesforce security configurations.
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Skills Students Will Acquire

- Salesforce Administration
- Salesforce Service Cloud Configuration
- Salesforce Data Modeling
- Custom Object and Field Creation
- Lookup Relationships
- Validation Rules
- Record-Triggered Flows
- Flow Automation
- Lightning App Builder

- Prompt Builder
- Agentforce Fundamentals
- Reports and Dashboard Creation
- Queue Management
- Salesforce Security Configuration
- Customer Support Process Automation
- AI-powered Business Process Automation



System Requirements

Operating System

- Windows 10 or later / Windows 11
- macOS
- Linux (Ubuntu or equivalent)

A 64-bit operating system is recommended for better performance while accessing Salesforce and supporting development tools.

Processor

- **Minimum:** Intel Core i3 (or equivalent AMD processor)
- **Recommended:** Intel Core i5 or higher

A higher-performance processor provides a smoother experience when working with Salesforce Lightning Experience and browser-based tools.

Memory (RAM)

- **Minimum:** 8 GB RAM
- **Recommended:** 16 GB RAM

Adequate memory ensures smooth multitasking while using Salesforce, Microsoft Word, GitHub, and web browsers.

Storage

- **Minimum:** 10 GB free disk space
 - **Recommended:** SSD storage for improved performance
-

Internet Connection

A stable broadband internet connection with a minimum speed of **5 Mbps** is recommended for accessing Salesforce Developer Edition, Trailhead, GitHub, and other cloud services.

Software Requirements

Software	Purpose
Salesforce Developer Edition	Salesforce development environment
Google Chrome / Microsoft Edge	Access Salesforce Lightning Experience
Microsoft Word	Documentation preparation
GitHub	Version control and project repository
Microsoft PowerPoint	Project presentation

Technologies Used

- Salesforce Service Cloud

- Lightning Experience
- Flow Builder
- Prompt Builder
- Lightning App Builder
- Agentforce
- Reports and Dashboards

Introduction to Salesforce

Salesforce is one of the world's leading cloud-based Customer Relationship Management (CRM) platforms. It enables organizations to manage customer relationships, automate business processes, and improve collaboration across departments. Salesforce provides a wide range of cloud services, including Sales Cloud, Service Cloud, Marketing Cloud, Commerce Cloud, and Experience Cloud, allowing businesses to manage different aspects of their operations from a single platform.

For customer support operations, Salesforce Service Cloud offers powerful features such as case management, automation, knowledge management, reporting, and AI-driven assistance. Organizations can use these features to provide faster customer service, improve issue resolution, and maintain consistent communication with customers.

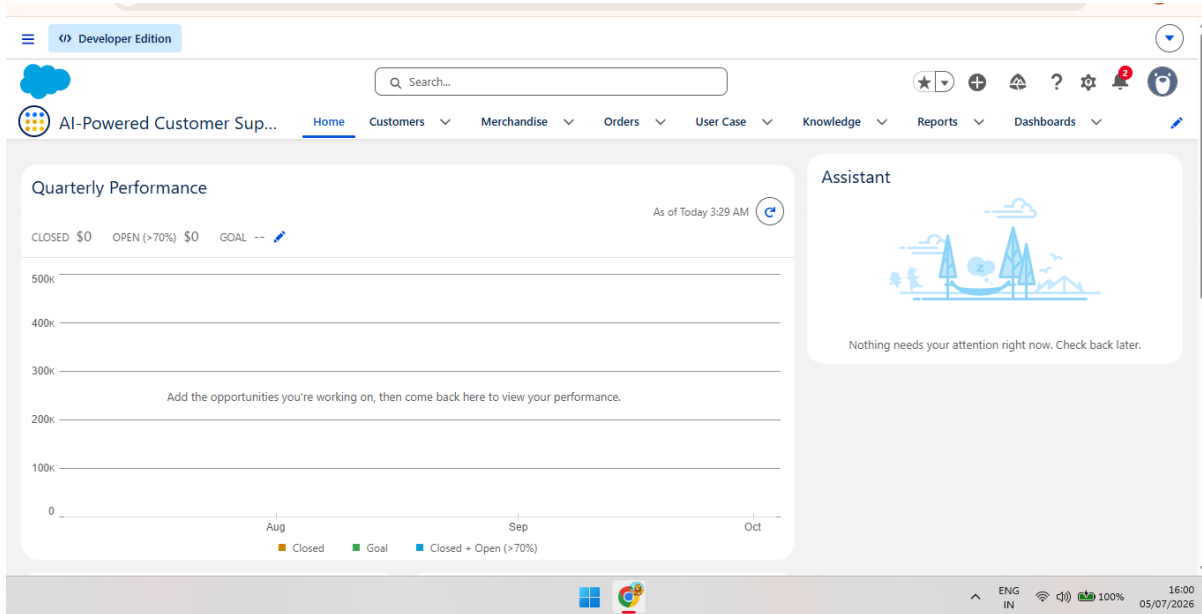
In this project, Salesforce Service Cloud serves as the foundation for developing an AI-powered customer support automation system. Custom objects are created to manage customers, merchandise, orders, user cases, and knowledge articles. Validation rules ensure that users enter accurate information, while Flow Builder automates repetitive processes such as updating case status and handling high-priority support requests.

Prompt Builder extends the solution by generating AI-powered summaries based on customer case details. These summaries help support agents understand customer issues more efficiently and maintain consistent documentation. Lightning App Builder is used to create a user-friendly interface that provides centralized access to all project components.

By integrating Salesforce automation with AI-powered capabilities, the project demonstrates how modern CRM technologies can simplify customer service operations, improve productivity, and enhance the overall customer experience.

Benefits of Salesforce in This Project

- Centralized customer and product management.
- Automated customer support workflows.
- Improved data quality using validation rules.
- Faster case handling through record-triggered flows.
- AI-assisted case summaries using Prompt Builder.
- Better visibility through reports and dashboards.
- Secure, scalable, cloud-based architecture.



Milestone 2 – Backend Development & Configuration

Introduction

The Backend Development and Configuration phase focuses on building the core data structure required for the AI-Powered Customer Support & Service Automation system. In Salesforce, custom objects, fields, relationships, and validation rules are configured to support customer service operations.

The objective of this phase is to create a structured database capable of managing customers, merchandise, orders, user cases, and knowledge articles while maintaining data consistency through validation rules and automation. Proper backend configuration ensures that all business information is organized efficiently and can be accessed securely by users.

This phase also establishes the foundation for Flow automation, Prompt Builder, and reporting features implemented in later stages of the project.

Activity 1 – Creating Salesforce Developer Account

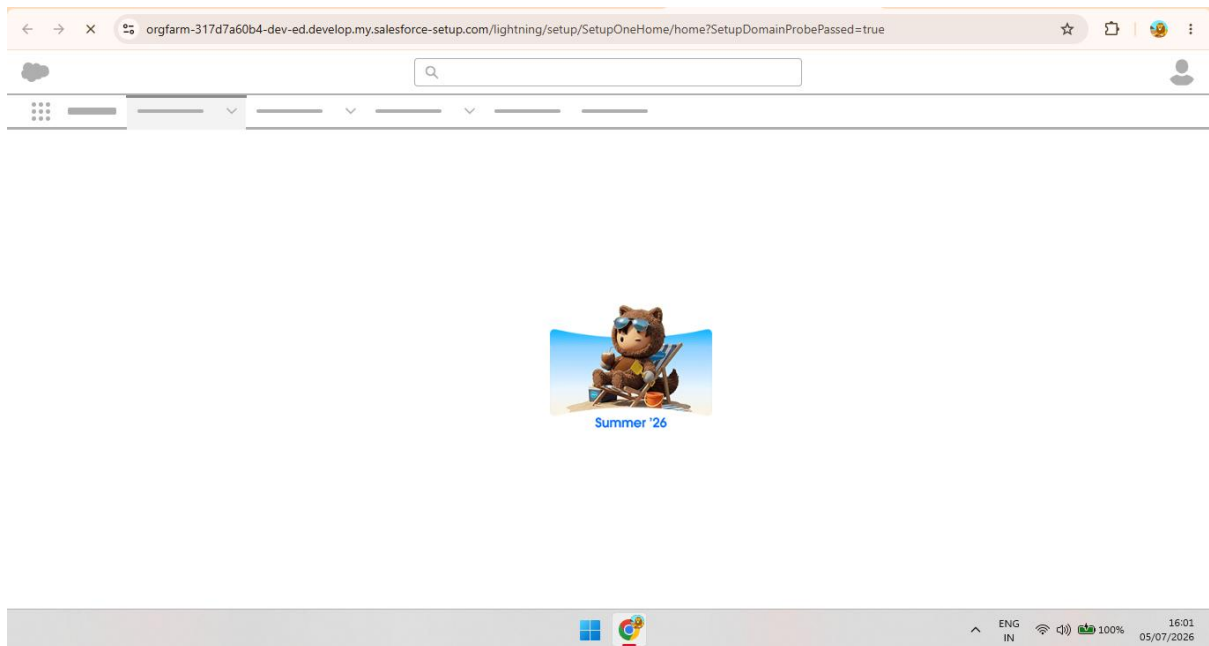
Objective

A Salesforce Developer Edition account provides a free cloud environment where developers can configure applications, create custom objects, automate business processes, and implement AI-powered features without affecting production environments.

Steps Performed

1. Navigate to the Salesforce Developer signup page.
2. Enter personal details including:
 - First Name
 - Last Name
 - Email Address
 - Role
 - Company Name
 - Country
 - Postal Code
3. Provide a unique username in email format.
4. Click **Sign Me Up**.
5. Verify the account using the activation email.
6. Create a secure password.
7. Log in to the Salesforce Developer Edition.

The Developer Edition provides access to Lightning Experience, Service Cloud features, Flow Builder, Prompt Builder, Lightning App Builder, and other Salesforce tools required for this project.



Activity 2 – Creating Custom Objects

Objective

Custom objects are created to store and organize business information related to customer support. Each object represents a specific business entity and contains fields that capture relevant information.

The project includes the following custom objects:

- Customer
- Merchandise
- Orders
- User Case
- Knowledge

These objects work together to manage customer interactions, product details, order information, support requests, and knowledge articles.

Customer Object

Purpose

The **Customer** object stores customer information required for managing support requests. It maintains contact details and customer status, allowing support agents to quickly identify and assist customers.

Customer Fields

Field Label	Data Type
Customer Name	Text
Email	Email
Phone	Phone
Customer Type	Picklist (Individual, Business)
Address	Long Text Area
Status	Picklist (Active, Inactive)

The Customer object serves as the primary record for customer information and is referenced by both Orders and User Case objects.

Customer | Salesforce

Ask Gemini

orgfarm-317d7a60b4-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lg50000034kf/FieldsAndRelationships/view

Developer Edition

Search Setup

SetupHomeObject Manager

Customer

DetailsFields & RelationshipsPage LayoutsLightning Record PagesButtons, Links, and ActionsCompact LayoutsField SetsObject LimitsRecord TypesRelated Lookup Filters

Fields & Relationships

9 Items, Sorted by Field Label

Quick Find

NewDeleted FieldsField DependenciesSet History Tracking

Address	Address__c	Long Text Area(32768)	
Created By	CreatedById	Lookup(User)	
Customer Name	Name	Text(80)	✓
Customer Type	Customer_Type__c	Picklist	
Email	Email__c	Email	
Last Modified By	LastModifiedById	Lookup(User)	
Owner	OwnerId	Lookup(User,Group)	

My Service Journey

Object Manager | Salesforce

Ask Gemini

orgfarm-317d7a60b4-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/home

Developer Edition

Search Setup

SetupHomeObject Manager

Object Manager

53+ Items, Sorted by Label

Quick Find

Schema BuilderCreate

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Account	Account	Standard Object			
Activity	Activity	Standard Object			
Activity History	DevopsActivityLog	Standard Object			
Address	Address	Standard Object			
Agent Work	AgentWork	Standard Object			
Alternative Payment Method	AlternativePaymentMethod	Standard Object			
API Anomaly Event Store	ApiAnomalyEventStore	Standard Object			
Appointment Category	AppointmentCategory	Standard Object			
Appointment Invitation	AppointmentInvitation	Standard Object			

My Service Journey

Merchandise Object

Purpose

The **Merchandise** object stores product information including warranty details, product category, and availability. This object enables support agents to identify products associated with customer orders and warranty claims.

Merchandise Fields

Field Label	Data Type
Product Name	Text
Product Code	Text
Category	Picklist
Warranty Months Number	
Active	Checkbox

Each merchandise record represents a product offered by the company.

Orders Object

Purpose

The **Orders** object stores purchase information for customers. It connects customers with the products they have purchased and provides information required for order tracking and warranty verification.

Orders Fields

Field Label	Data Type
Order Number	Auto Number
Order Date	Date
Delivery Date	Date
Status	Picklist (New, Shipped, Delivered)
Customer	Lookup(Customer)
Merchandise	Lookup(Merchandise)

Lookup relationships enable the Orders object to associate each order with a specific customer and merchandise item.

User Case Object

Purpose

The **User Case** object is the core component of the application. It stores customer complaints, warranty requests, delivery issues, and product-related support cases. This object supports automation through validation rules, flows, and AI-generated summaries.

User Case Fields

Field Label Data Type

Case Number	Auto Number
Customer	Lookup(Customer)
Merchandise	Lookup(Merchandise)
Orders	Lookup(Orders)
Case Type	Picklist
Priority	Picklist
Status	Picklist
Description	Long Text Area
AI Summary	Long Text Area

The User Case object integrates with Prompt Builder to generate AI summaries based on case information.

Knowledge Object

Purpose

The **Knowledge** object is designed to store articles, frequently asked questions (FAQs), troubleshooting guides, warranty policies, and other support-related information. These knowledge articles help customer support representatives provide accurate and consistent responses to customers while reducing the time required to resolve common issues.

The Knowledge object also supports future enhancements such as AI-assisted recommendations and self-service portals where customers can access solutions without contacting the support team.

Custom Fields

Field Label Data Type

Article Title Text

Category Picklist

Content Long Text Area

Active Checkbox

The Active field determines whether a knowledge article is currently available for customer support representatives.

Steps to Create Knowledge Object

1. Navigate to **Setup**.
2. Open **Object Manager**.
3. Click **Create → Custom Object**.
4. Enter the following information:
 - Label: Knowledge
 - Plural Label: Knowledge
 - Record Name: Article Title
5. Enable:
 - Allow Reports
 - Allow Search
6. Save the object.
7. Create all required custom fields.
8. Assign the object to the Lightning Application.

Tabs Creation

Purpose

Tabs provide quick access to Salesforce objects through the Lightning Application. Creating tabs improves navigation by allowing users to directly open Customer, Merchandise, Orders, User Case, and Knowledge records without navigating through Setup.

The project includes custom tabs for all custom objects to provide a centralized and user-friendly interface.

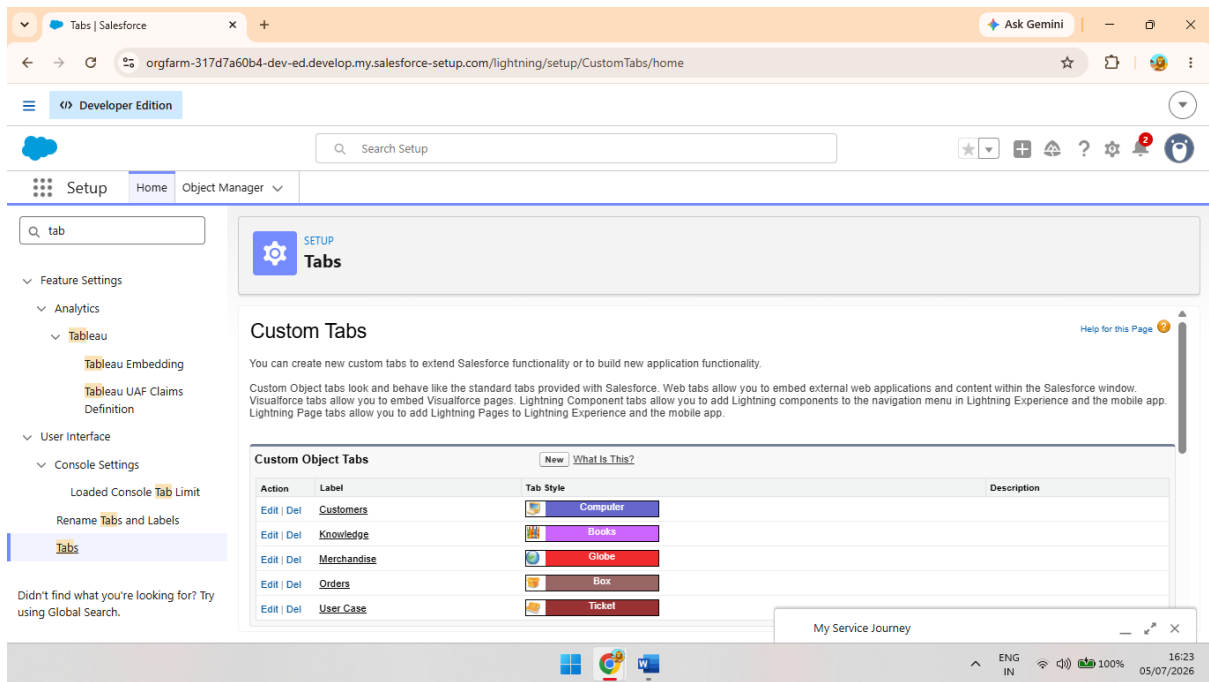
Steps to Create Tabs

1. Navigate to **Setup**.
2. Search for **Tabs**.
3. Select **Custom Object Tabs**.
4. Click **New**.
5. Choose the required object.
6. Select a suitable tab style.
7. Assign the tab to all required profiles.
8. Save the configuration.

The same procedure was followed to create tabs for all custom objects developed in this project.

Tabs Created

- Customer
- Merchandise
- Orders
- User Case
- Knowledge



Object Relationships

Purpose

Relationships are essential in Salesforce because they connect business information across different objects. They allow users to retrieve related records quickly and maintain data consistency throughout the application.

The AI-Powered Customer Support & Service Automation project uses Lookup Relationships to establish connections between customers, products, orders, and support cases.

Relationships Implemented

Customer → Orders

One customer can place multiple orders.

Relationship Type:

Lookup Relationship

Merchandise → Orders

Each order references one merchandise item.

Relationship Type:

Lookup Relationship

Customer → User Case

A customer may create multiple support cases.

Relationship Type:

Lookup Relationship

Merchandise → User Case

Each support case references the related merchandise item.

Relationship Type:

Lookup Relationship

Orders → User Case

Support cases are linked to customer orders for warranty verification and order tracking.

Relationship Type:

Lookup Relationship

Data Model

The data model provides a structured representation of the application's database. Each object stores a specific category of business information while lookup relationships connect related records.

The Customer object serves as the central repository for customer information. Merchandise stores product details, while Orders links customers to purchased products. User Case records maintain customer complaints, warranty claims, and support requests. Knowledge stores support articles and frequently used solutions.

This structured data model enables efficient reporting, automation, and AI-powered processing within Salesforce.

Milestone 3 – Validation Rules & Process Automation

Introduction

Validation rules and process automation are essential components of Salesforce applications because they ensure that business data remains accurate, complete, and

consistent. Validation rules prevent users from saving records that violate business requirements, while Flow Builder automates repetitive tasks without requiring manual intervention.

In the AI-Powered Customer Support & Service Automation project, validation rules were implemented on the **User Case** object to enforce mandatory business requirements before support cases are created. Record-triggered flows were also configured to automate case status updates and case assignment based on predefined business logic.

These automations improve data quality, reduce manual effort, and ensure that customer support requests are processed efficiently.

Activity 1 – Validation Rules

Purpose

Validation rules ensure that customer support cases contain all mandatory information before they are saved. This prevents incomplete or incorrect records from entering the system and helps maintain consistent business data.

The project includes two validation rules implemented on the **User Case** object.

Validation Rule 1 – Description Required

Objective

Every customer support case should contain a detailed description of the reported issue. Without a description, support agents may not have sufficient information to investigate or resolve the problem.

This validation rule prevents users from saving a User Case if the Description field is left blank.

Validation Formula

ISBLANK(Description__c)

Error Message

Please enter the case description.

Error Location

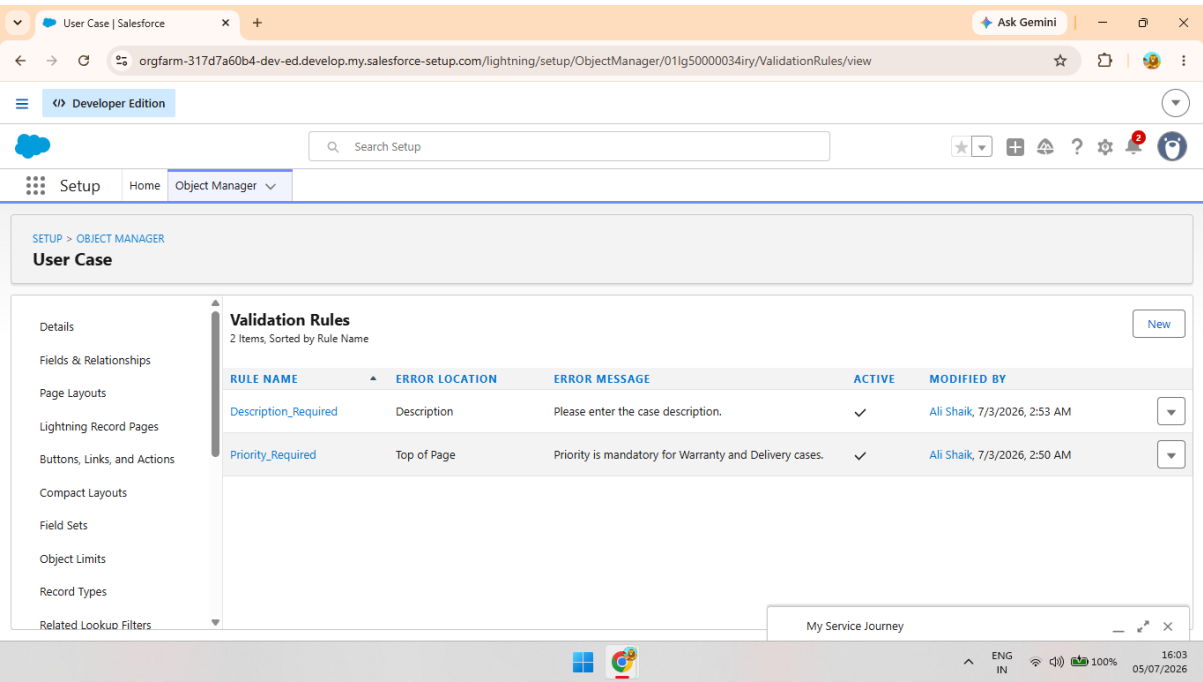
Description Field

Implementation Steps

- 1. Navigate to **Setup**.
- 2. Open **Object Manager**.
- 3. Select **User Case**.
- 4. Click **Validation Rules**.
- 5. Click **New**.
- 6. Enter the rule name:
 - **Description_Required**
- 7. Enter the validation formula.
- 8. Specify the error message.
- 9. Save and Activate the validation rule.

Expected Result

If a user attempts to save a User Case without entering a description, Salesforce displays an error message and prevents the record from being saved.



Validation Rule 2 – Priority Required

Objective

Warranty and Delivery cases require immediate attention. Therefore, the Priority field must always be specified whenever the Case Type is either **Warranty** or **Delivery**.

This validation rule ensures that users cannot save these cases without selecting an appropriate priority.

Validation Formula

```
AND(  
OR(  
ISPICKVAL(Case_Type__c,"Warranty"),  
ISPICKVAL(Case_Type__c,"Delivery")  
),  
ISBLANK(TEXT(Priority__c))  
)
```

Error Message

Priority is mandatory for Warranty and Delivery cases.

Error Location

Top of Page

Implementation Steps

1. Navigate to **Setup**.
2. Open **Object Manager**.
3. Select **User Case**.
4. Open **Validation Rules**.
5. Click **New**.
6. Enter:

Rule Name

Priority_Required

7. Paste the validation formula.
8. Enter the error message.
9. Save and Activate the validation rule.

Expected Result

If the Case Type is Warranty or Delivery and Priority is not selected, Salesforce prevents the record from being saved.

Benefits of Validation Rules

The implemented validation rules provide several business benefits:

- Prevent incomplete customer support cases.
- Improve data quality.
- Ensure mandatory information is captured.
- Reduce manual verification by support agents.
- Maintain consistency across all customer support records.

Activity 2 – Flow Automation

Purpose

Flow Builder enables Salesforce administrators to automate business processes without writing code. Record-triggered flows execute automatically when records are created or updated, reducing manual work and ensuring consistent business operations.

Two record-triggered flows were implemented as part of this project.

Flow 1 – Default Case Status Flow

Objective

Whenever a new User Case is created, the system automatically assigns a default status of **New**. This ensures that all newly created cases begin with a consistent status and are ready for processing by the support team.

Flow Type

Record-Triggered Flow

Trigger

Object:

User Case

Trigger Condition:

A record is created

Flow Logic

When a new User Case is created:

↓

Automatically update

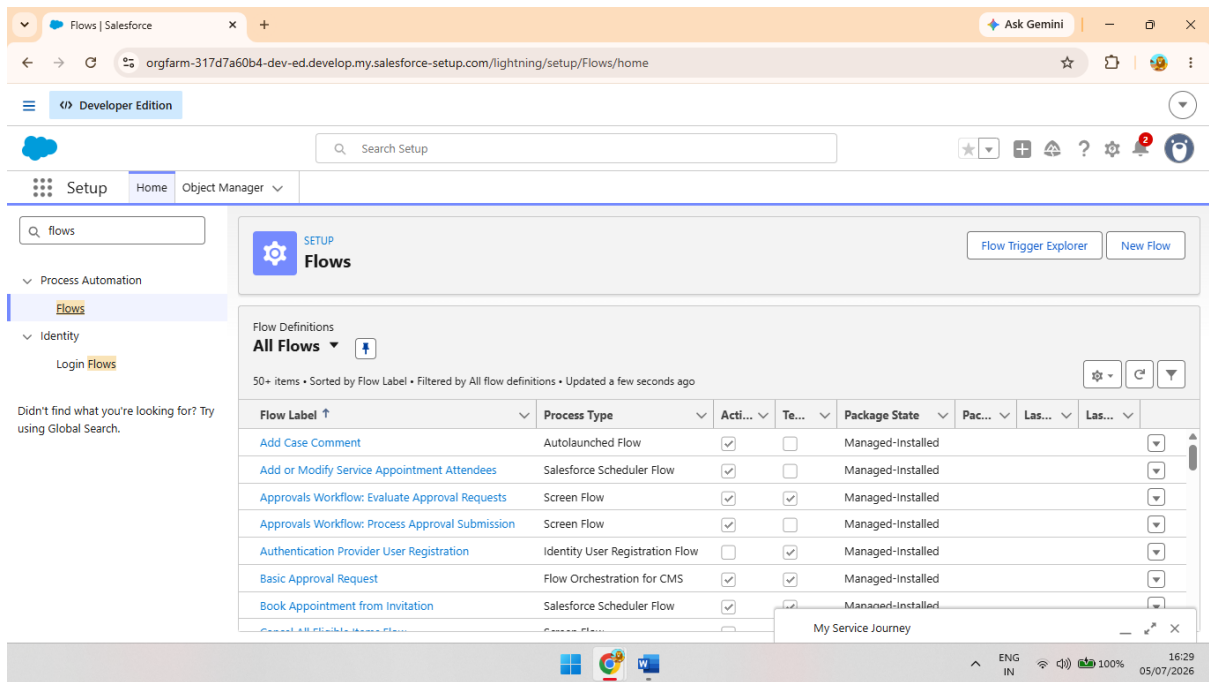
Status = New

↓

Save the record

Business Benefits

- Eliminates manual status updates.
- Standardizes the customer support process.
- Ensures all new cases enter the support workflow correctly.



Flow 2 – High Priority Case Escalation

Objective

Customer support teams must respond quickly to high-priority issues such as warranty claims, damaged products, or urgent delivery problems. To reduce manual effort and ensure immediate attention, a Record-Triggered Flow was created to automatically update high-priority cases.

Whenever a User Case is created with **Priority = High**, Salesforce automatically updates the case status to **Escalated**, allowing support teams to identify urgent requests without manually reviewing every case.

Flow Type

Record-Triggered Flow

Trigger Configuration

Property	Value
Object	User Case
Trigger	A record is created

Optimize For Actions and Related Records

Flow Logic

User Case Created



Check Priority



Priority = High ?

/ \

Yes No



Update Status

Status = Escalated



Save Record

Implementation Steps

1. Navigate to **Setup**.
2. Search for **Flows**.
3. Click **New Flow**.
4. Select **Record-Triggered Flow**.
5. Choose **User Case** as the object.
6. Configure the trigger to run **when a record is created**.
7. Add a **Decision** element named **Is High Priority**.
8. Configure the decision:
 - Field: Priority
 - Operator: Equals
 - Value: High
9. Add an **Update Records** element.
10. Update the Status field to **Escalated**.

11. Save the Flow.

12. Activate the Flow.

Expected Result

Whenever a User Case with **Priority = High** is created, Salesforce automatically changes the Status to **Escalated**.

This ensures urgent customer requests are immediately visible to the support team.

Business Benefits

- Faster handling of critical customer issues.
- Automatic escalation without manual intervention.
- Improved customer satisfaction.
- Consistent support workflow.

Activity 3 – Queue Configuration

Purpose

Queues help distribute customer support cases among different support teams. Instead of assigning every case to a single user, Salesforce automatically places records into a queue where authorized team members can access and manage them.

This improves workload distribution and ensures that customer requests are handled by the appropriate support specialists.

Warranty Support Queue

Objective

The Warranty Support Queue is responsible for handling warranty-related customer issues.

Whenever warranty cases are created, they can be assigned to this queue for efficient processing.

Queue Configuration

Property	Value
Queue Label	Warranty Support Queue
Queue Name	Warranty_Support_Queue
Supported Object	User Case

Implementation Steps

1. Navigate to **Setup**.
2. Search for **Queues**.
3. Click **New**.
4. Enter the Queue Label:
Warranty Support Queue
5. Select **User Case** under Supported Objects.
6. Add queue members.
7. Save.

Product Support Queue

Objective

The Product Support Queue is responsible for handling customer complaints related to damaged or defective merchandise.

Queue Configuration

Property	Value
Queue Label	Product Support Queue
Queue Name	Product_Support_Queue
Supported Object	User Case

Implementation Steps

1. Open **Setup**.

2. Search for **Queues**.
3. Click **New**.
4. Enter Queue Name.
5. Select **User Case**.
6. Assign members.
7. Save.

Product Support Queue

Objective

The Product Support Queue is responsible for handling customer complaints related to damaged or defective merchandise.

Queue Configuration

Property	Value
Queue Label	Product Support Queue
Queue Name	Product_Support_Queue
Supported Object	User Case

Implementation Steps

1. Open **Setup**.
2. Search for **Queues**.
3. Click **New**.
4. Enter Queue Name.
5. Select **User Case**.
6. Assign members.
7. Save.

Milestone 5 – Agentforce, Reporting & Deployment

Introduction

Artificial Intelligence has become an essential part of modern customer service by enabling organizations to provide faster, more accurate, and consistent support. Salesforce Agentforce and Prompt Builder extend Salesforce capabilities by generating AI-powered content based on business data stored within Salesforce objects.

In this project, Salesforce Prompt Builder was implemented to automatically generate customer support summaries for User Case records. Instead of manually writing summaries for every customer issue, AI analyzes the available case information and creates a professional summary that helps support representatives quickly understand the customer's problem.

This milestone also covers Prompt Builder deployment, testing, reports, dashboards, system deployment, maintenance, and future enhancements.

Activity 1 – Creating Prompt Builder Template

Purpose

Prompt Builder enables Salesforce administrators to create AI-powered prompt templates without writing code. These templates use Salesforce data to generate intelligent responses and summaries.

For this project, a Field Generation Prompt Template was created for the **User Case** object to generate professional summaries based on customer support information.

Prerequisites

Before creating the Prompt Template, the **AI Summary** field must exist on the User Case object.

Field Details:

Property	Value
Field Label	AI Summary
Data Type	Long Text Area
Length	32000
Visible Lines	5

Steps to Create Prompt Builder

1. Navigate to **Setup**.
2. Search for **Einstein Setup**.
3. Enable Einstein AI features.
4. Refresh the page.
5. Search for **Prompt Builder**.
6. Click **New Prompt Template**.
7. Select **Field Generation**.
8. Click **Next**.

Prompt Template Configuration

Property	Value
Template Name	User Case AI Summary
Prompt Type	Field Generation
Object	User Case
Target Field	AI Summary

Resources Used

The following User Case fields were included:

- Case Number
- Customer
- Merchandise
- Orders
- Case Type
- Priority
- Status
- Description

These fields provide sufficient information for the AI model to generate an accurate summary.

Prompt Used

Generate a professional customer support summary using the provided User Case information.

Include:

- Customer issue
- Product involved
- Case type
- Priority level
- Current status
- Recommended support action

Write the summary in a clear, professional, and business-friendly manner suitable for customer support representatives.

Expected Output

Example:

Customer Rahul Sharma reported that the Dell Inspiron 15 laptop screen flickers during startup. The issue has been classified as a Warranty case with High priority and has been automatically escalated for immediate technical review. The product remains under warranty, and the support team should coordinate hardware diagnostics and replacement if necessary.

Deploying Prompt Builder

Purpose

After creating the Prompt Template, it must be assigned to the AI Summary field so that users can generate summaries directly from User Case records.

Steps

1. Open any User Case record.
2. Click **Gear** → **Edit Page**.
3. Upgrade to Dynamic Forms if required.
4. Select the **AI Summary** field.
5. Assign the Prompt Template.
6. Save.
7. Activate the Lightning Page.
8. Assign as Org Default.

Testing AI Summary

Purpose

Testing verifies that Prompt Builder generates summaries correctly using User Case information.

Test Scenario

Customer:

Rahul Sharma

Product:

Dell Inspiron 15

Case Type:

Warranty

Priority:

High

Description:

Laptop screen flickers after startup.

Expected Result

The AI Summary field should automatically generate a professional summary describing the customer's issue, current case status, and recommended action.

Reports

Purpose

Reports help managers monitor customer support operations and analyze business performance.

The following reports were created:

User Cases by Status

Displays the number of support cases grouped by their current status.

User Cases by Priority

Shows the number of High, Medium, and Low priority cases.

Orders by Status

Displays customer orders categorized by New, Shipped, and Delivered.

Customer Report

Lists all registered customers.

Merchandise Report

Displays available products and warranty information.

Dashboard

Purpose

Dashboards provide graphical visualization of customer support activities.

The Customer Support Dashboard includes:

- Cases by Status
- Cases by Priority

- Orders by Status
- Customer Distribution
- Product Categories

The dashboard enables management to monitor support performance through charts and graphs.

Deployment

Purpose

Deployment transfers Salesforce configurations from the development environment to a production-ready environment.

The following components were successfully configured and prepared for deployment:

- Custom Objects
- Custom Fields
- Validation Rules
- Record-Triggered Flows
- Support Queues
- Lightning Application
- Prompt Builder Template
- Reports
- Dashboard

Deployment can be performed using Salesforce Change Sets or Salesforce CLI after successful testing.

Maintenance

Regular maintenance ensures the application continues to operate efficiently.

Maintenance activities include:

- Monitoring Flow executions.
- Updating Prompt Templates.
- Reviewing Validation Rules.
- Maintaining support queues.

- Updating reports and dashboards.
 - Applying Salesforce platform updates.
 - Monitoring system performance.
-

Future Enhancements

The current system can be extended by implementing additional Salesforce capabilities, including:

- AI Chatbot Integration.
 - Email-to-Case.
 - Customer Self-Service Portal.
 - Predictive Case Routing.
 - Service Level Agreement (SLA) Monitoring.
 - Omni-Channel Support.
 - Mobile Salesforce Application.
 - Real-Time Notifications.
 - Integration with external ERP systems.
 - AI-based sentiment analysis for customer complaints.
-

Conclusion

The AI-Powered Customer Support & Service Automation project successfully demonstrates the implementation of Salesforce Service Cloud to automate customer support operations. The solution integrates custom objects, validation rules, record-triggered flows, support queues, Lightning Applications, and Prompt Builder to streamline customer service processes.

By automating repetitive tasks and generating AI-powered summaries, the system reduces manual effort, improves response time, enhances data accuracy, and increases overall customer satisfaction. The project also provides practical experience in Salesforce administration, automation, and AI technologies while demonstrating how modern CRM platforms can support enterprise customer service operations.

The completed solution is scalable, secure, and capable of supporting future enhancements as organizational requirements evolve.

References

1. Salesforce Developer Documentation
2. Salesforce Trailhead Learning Platform
3. Salesforce Service Cloud Documentation
4. Salesforce Flow Builder Guide
5. Salesforce Prompt Builder Documentation
6. Salesforce Agentforce Documentation
7. APSCHE Salesforce Virtual Internship Project Guidelines